

ISyE 6669 Final

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Problem 1. Which of the following sets is convex for any positive integer n ?

A $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \geq 1\}.$

B $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i = 1\}.$

C $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \leq 1\}.$

Solution: C

Problem 2. Which of the following functions is convex on the specified domain?

A $f(x) = 1 - |x|$ on $[-1, 1] \subseteq \mathbb{R}.$

B $f(x) = |x|$ on $X = \{x \in \mathbb{R} : |x| \geq 1\}.$

C $f(x) = -|x|$ on $X = \{x \in \mathbb{R} : |x| \leq 0\}.$

Solution: C

Problem 3. Which of the following statements is true about the extreme points and extreme rays?

A Any polyhedron has an extreme point.

B If a polyhedron does not have any extreme ray, then it is bounded.

C For a polyhedron, any point in the polyhedron can be written as the sum of some extreme points and some extreme rays.

Solution: B

Problem 4. Consider a constrained optimization problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g(\mathbf{x}) \leq 0, \\ & h(\mathbf{x}) = 0, \\ & \mathbf{x} \in X. \end{aligned}$$

Which of the following problems is a relaxation of this problem?

A $\min\{f(\mathbf{x}) + h(\mathbf{x}) : g(\mathbf{x}) \leq 0, \mathbf{x} \in X\}.$

B $\max\{f(\mathbf{x}) + g(\mathbf{x}) : h(\mathbf{x}) = 0, \mathbf{x} \in X\}.$

C $\min\{h(\mathbf{x}) + g(\mathbf{x}) : \mathbf{x} \in X\}.$

Solution: A

Problem 5. Consider the following linear program for $(x_1, x_2, x_3) \in \mathbb{R}^3$:

$$\begin{aligned} \min \quad & x_1 + 2x_2 + 3x_3 \\ \text{s.t.} \quad & -x_1 + 3x_2 = 5, \\ & 2x_1 - x_2 + 3x_3 \geq 6, \\ & x_3 \leq 4, \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.} \end{aligned}$$

The dual of the above linear program can be written as:

A

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 4y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \leq 1, \\ & 3y_1 - y_2 \geq 2, \\ & 3y_2 + y_3 = 3, \\ & y_1 \text{ free}, y_2 \leq 0, y_3 \geq 0. \end{aligned}$$

B

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 4y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \leq 1, \\ & 3y_1 - y_2 \geq 2, \\ & 3y_2 + y_3 = 3, \\ & y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0. \end{aligned}$$

C

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 4y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \geq 1, \\ & 3y_1 - y_2 \leq 2, \\ & 3y_2 + y_3 = 3, \\ & y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0. \end{aligned}$$

Solution: B

Problem 6. Consider the following linear program:

$$(P) \begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b, \\ & 0 \leq x \leq d, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. Which of the following statements are true for any data A , b , and c ?

- A The problem (P) is always feasible.
- B If the problem (P) is infeasible, then its dual is unbounded.
- C The problem (P) and its dual can be infeasible at the same time.

Solution: B

Problem 7. The robust linear constraint for $x \in \mathbb{R}^n$:

$$a^\top x \geq b, \forall a \in U = \{a \in [0, 1]^n : Ca \leq d\}$$

is equivalent to

- A $\sum_{i=1}^n x_i \geq b$.
- B $C^\top x - d \geq b$.
- C $d^\top y + e^\top z + b \leq 0$, $x + C^\top y + z \geq 0$, $y, z \geq 0$, where $e = (1, \dots, 1)^\top \in \mathbb{R}^n$.

Solution: C

Problem 8. Consider a simple cutting stock problem, where a big roll has width 7 and demand takes widths 2 and 3. How many patterns are there for cutting a big roll into smaller widths of the demand?

- A 5
- B 6
- C 7

Solution: C

Problem 9. Which of the following statements is NOT true?

- A The column generation algorithm will terminate if all the columns are added.

- B The constraint generation algorithm will terminate if all the constraints are added.
- C The constraint generation algorithm will not terminate before all the constraints are added.

Solution: C

Problem 10. For the polytope $P = [0, 1]^n$, the extreme point representation of $x \in P$ can be written as

- A $x = \sum_{i=1}^n \lambda_i e_i$, $\sum_{i=1}^n \lambda_i = 1$, $\lambda_i \geq 0$, $i = 1, \dots, n$, where e_i is the unit vector with i th component being 1.
- B $x = \sum_{i=1}^n \lambda_i e_i$, $\sum_{i=1}^n \lambda_i \leq 1$, $\lambda_i \geq 0$, $i = 1, \dots, n$, where e_i is the unit vector with i th component being 1.
- C $x = \sum_{i=1}^n \lambda_i e_i$, $\sum_{i=1}^n \lambda_i \geq 1$, $\lambda_i \geq 0$, $i = 1, \dots, n$, where e_i is the unit vector with i th component being 1.

Solution: B

Problem 11. Consider the following linear program:

$$\begin{aligned}
 \min \quad & 3x_1 + 2x_2 + x_3 + 2x_4 & (1) \\
 \text{s.t.} \quad & x_1 - 3x_2 + 2x_3 + x_4 \leq 1 & (2) \\
 & x_1 + x_2 \leq 1 & (3) \\
 & -x_1 + x_2 \leq 1 & (4) \\
 & x_2 \geq 0 & (5) \\
 & 0 \leq x_3 \leq 1 & (6) \\
 & 0 \leq x_4 \leq 1 & (7)
 \end{aligned}$$

Denote the polyhedron for variables x_1 and x_2 defined by (3)-(5) as P_1 , and P_2 as the polyhedron for x_3 and x_4 defined by (6)-(7). Use the extreme point of P_1 : $(x_1^1, x_2^1) = (0, 1)$, and two extreme points of P_2 : $(x_3^1, x_4^1) = (0, 0)$ and $(x_3^2, x_4^2) = (1, 0)$. Which of the following is a correct restricted master problem of the linear program?

A

$$\begin{aligned}
 \min \quad & 2\lambda_1 + \beta_2 \\
 \text{s.t.} \quad & -3\lambda_1 + 2\beta_2 \leq 1 \\
 & \lambda_1 = 1 \\
 & \beta_1 + \beta_2 = 1 \\
 & \lambda_1 \geq 0 \\
 & \beta_1, \beta_2 \geq 0.
 \end{aligned}$$

B

$$\begin{array}{ll}\min & \lambda_1 + 2\beta_2 \\ \text{s.t.} & -2\lambda_1 + 3\beta_2 \leq 1 \\ & \lambda_1 = 1 \\ & \beta_1 + \beta_2 = 1 \\ & \lambda_1 \geq 0 \\ & \beta_1, \beta_2 \geq 0.\end{array}$$

C

$$\begin{array}{ll}\min & 2\lambda_1 + \beta_2 \\ \text{s.t.} & -2\lambda_1 + 3\beta_2 \leq 1 \\ & \lambda_1 = 1 \\ & \beta_1 + \beta_2 = 1 \\ & \lambda_1 \geq 0 \\ & \beta_1, \beta_2 \geq 0.\end{array}$$

Solution: A

Problem 12. Which of the following points is the closest to $(1, 2, 3, 4)^\top \in \mathbb{R}^4$ in Euclidean norm?

A $(3, 2, 1, 4)^\top$

B $(3, 1, 2, 4)^\top$

C $(4, 2, 1, 3)^\top$

Solution: B

Problem 13. If we want to solve a least square problem $\min_x \|Ax - b\|_2^2$ where $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ and

$b = (2, 3, 1)^\top$, then

A the Moore-Penrose pseudoinverse of A is $A^\dagger = \begin{bmatrix} \frac{1}{2} & -\frac{3}{2} & 0 \\ 1 & 1 & 0 \end{bmatrix}$;

B the optimal solution $x = (-\frac{7}{2}, 3)^\top$;

C the optimal solution $x = (3, -\frac{7}{2})^\top$.

Solution: B

Problem 14. If $A \in \mathbb{R}^{3 \times 3}$ is a symmetric matrix with singular values $\sigma_1 = 2$, $\sigma_2 = 1$, and $\sigma_3 = 0$, which of the following statements is true about the eigenvalues of A ?

- A The matrix A may not have eigenvalues.
- B The eigenvalues of A can be $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = 0$.
- C The eigenvalues of A can be $\lambda_1 = \sqrt{2}$, $\lambda_2 = 1$, and $\lambda_3 = 0$.

Solution: B

Problem 15. Consider the least square regression problem

$$z_2 = \min_{\beta_0, \dots, \beta_n} \sum_{i=1}^N (y_i - \sum_{j=1}^n \beta_j x_{ij})^2,$$

and the absolute deviation regression problem defined as

$$z_1 = \min_{\beta_0, \dots, \beta_n} \sum_{i=1}^N |y_i - \sum_{j=1}^n \beta_j x_{ij}|,$$

where $x_{ij}, i = 1, \dots, N, j = 1, \dots, n, y_i, i = 1, \dots, N$ are training data. Which of the following statements is true?

- A z_1 and z_2 are not equal, but the optimal solutions are always the same.
- B We cannot solve the absolute deviation regression problem by linear programming due to the nonlinear objective function.
- C The optimal solutions can be different.

Solution: C

Problem 16. Is $\begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix} \preceq_{\mathbb{R}_+^3} \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$?

- A Yes
- B No

Solution: B

Problem 17. Which of the following CANNOT be formulated as a SOC constraint?

A $x^2 + y^2 - 2x + 4y \leq 1$

B $x^2 - y^2 - 2x + 4y \leq 1$

C $xy - z^2 \geq 0$

D $x^2 + y^2 \geq 1$

Solution: D

Problem 18. Which of the following matrices is positive semidefinite?

A $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

B $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$

C $\begin{bmatrix} -3 & 2 \\ 3 & -3 \end{bmatrix}$

D $\begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$

Solution: D

Problem 19. Given a hyperplane $H : x_1 + 2x_2 = 2$ and a point $p = [1, 2]^\top$, find the distance d between H and p .

A $d = 4/5$

B $d = 3\sqrt{5}/5$

C $d = 2\sqrt{5}/5$

Solution: B

Problem 20. How many possible values can an integer variable x take where $x \in [0, 2]^2 \cap \mathbb{Z}^n$?

A 9

B 8

C 6

D 4

Solution: A

Problem 21. "*Either item 1 is not selected or item 2 is not selected*" can be modeled using binary variables $x_1, x_2 \in \{0, 1\}$ as:

A $x_1 \leq 1 - x_2$

B $1 - x_1 \leq x_2$

C $x_1 \leq x_2$

D $x_1 + x_2 \leq 1$

Solution: D

Problem 22. "*If item 1 is selected, then item 2 is selected.*" can be modeled using binary variables $x_1, x_2 \in \{0, 1\}$ as:

A $x_1 \leq 1 - x_2$

B $1 - x_1 \leq x_2$

C $x_1 \leq x_2$

D $x_1 + x_2 \leq 1$

Solution: C

Problem 23. "*If item 1 is selected, then item 2 and item 3 are selected.*" can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_1 \leq x_2 \leq x_3$

B $x_1 + x_2 \leq 1, x_1 + x_3 \leq 1$

C $x_1 \leq x_2, x_1 \leq x_3$

D $x_1 \leq x_2 + x_3$

Solution: C

Problem 24. "*If item 1 is selected, then item 2 or item 3 is selected.*" can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_1 \leq x_2 \leq x_3$

B $x_1 + x_2 \leq 1, x_1 + x_3 \leq 1$

C $x_1 \leq x_2, x_1 \leq x_3$

D $x_1 \leq x_2 + x_3$

Solution: D

Problem 25. "Either $x_1 + x_2 \leq 1$ or $x_1 - x_2 \geq 1$ " can be formulated as: $y \in \{0, 1\}$, $x_1 + x_2 \leq 1 + My$, and

A $x_1 - x_2 \geq 1 - M(1 - y)$

B $x_1 + x_2 \geq 1 + M(1 - y)$

C $x_1 - x_2 \leq 1 - M(1 - y)$

Solution: A

Problem 26. "If $x_1 + x_2 > 5$, then $x_1 - 2x_2 + x_3 \geq 8$ " can be formulated as: $y \in \{0, 1\}$, $x_1 - 2x_2 + x_3 \geq 8 - M(1 - y)$ and

A $x_1 + x_2 \geq 5 - M(1 - y)$

B $x_1 + x_2 \leq 5 + My$

C $x_1 + x_2 > 5 - M(1 - y)$

Solution: B

Problem 27. Consider the assignment problem

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = 1, \quad j = 1, \dots, n, \\ & \sum_{j=1}^n x_{ij} \leq 1, \quad i = 1, \dots, m, \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n, \end{aligned}$$

where i is the index for workers and j is the index for jobs. If we are going to model this problem as a network flow problem, how many nodes are supply nodes and how many nodes are demand nodes?

- A m supply nodes and n demand nodes
- B n supply nodes and m demand nodes
- C m supply nodes and $n + 1$ demand nodes
- D n supply nodes and $m + 1$ demand nodes

Solution: C

Problem 28. Consider an integer programming problem and its LP relaxation

$$\begin{aligned}
 z^{IP} = \min \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & x \in \{0, 1\}^n, \\
 z^{LP} = \min \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & x \in [0, 1]^n.
 \end{aligned}$$

Assume we know that $Z^{LP} = 10$. Then

- A $Z^{IP} = 10$.
- B it is possible that $Z^{IP} < 10$ for some particular c, A, b
- C $Z^{IP} \geq 10$

Solution: C

Problem 29. Consider an integer programming problem and its LP relaxation

$$\begin{aligned}
 z^{IP} = \min \quad & c^\top x \\
 \text{s.t.} \quad & \sum_{i=1}^n x_i \leq 1 \\
 & x \in \{0, 1\}^n, \\
 z^{LP} = \min \quad & c^\top x \\
 \text{s.t.} \quad & \sum_{i=1}^n x_i \leq 1 \\
 & x \in [0, 1]^n.
 \end{aligned}$$

Assume Z^{IP} and Z^{LP} are finite, then which of the following must be true?

- A $Z^{LP} < Z^{IP}$

B $Z^{LP} = Z^{IP}$

C $Z^{LP} > Z^{IP}$

Solution: B

Problem 30. In the course of a branch-and-bound algorithm for a maximization integer program, we have the following nodes (indicating the LP relaxation value and whether the LP solution is fractional or integer):

- Node 1: $V^{LP} = 9.9$; fractional solution
- Node 2: $V^{LP} = 10$; integer solution
- Node 3: $V^{LP} = 10.5$; fractional solution
- Node 4: $V^{LP} = 11$; fractional solution
- Node 5: $V^{LP} = 10.3$; integer solution

Which nodes need to be further explored?

A Node 1

B Node 2 and Node 5

C Node 3 and Node 5

D Node 3 and Node 4

Solution: D